

Divergence theorem of Gauss

(Transformation b/w triple integral and surface integral)

Statement

Let T be a closed bounded region in space boundary is a piecewise smooth oriented surface S . Let $\vec{F} = P(x,y,z)\hat{i} + Q(x,y,z)\hat{j} + R(x,y,z)\hat{k}$ — (1)

be a vector function, then

$$\boxed{\int \int \int_T (\text{div } \vec{F}) \, dv = \int \int_S (\vec{F} \cdot \hat{n}) \, ds} \quad \text{--- (2)}$$

(triple integral) (surface integral)

Q2

$$\boxed{\int \int \int_T \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} \right) dx \, dy \, dz = \int \int_S P \, dy \, dz + Q \, dz \, dx + R \, dx \, dy} \quad \text{--- (3)}$$

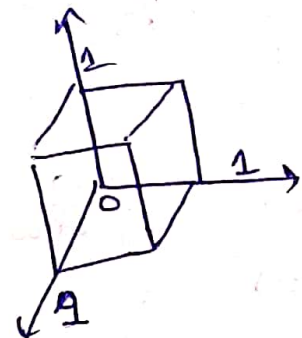
Ex

Find flux (surface integral) of

$$\vec{F} = \underset{\substack{\downarrow P \\ (1)}}{2xy} \hat{i} + \underset{\substack{\downarrow Q \\ (1)}}{3ye^z} \hat{j} + \underset{\substack{\downarrow R \\ (1)}}{(x \sin z)} \hat{k}$$

outward through the surface of cube by using Gauss divergence theorem.

$$\begin{aligned} 0 &\leq x \leq 1 \\ 0 &\leq y \leq 1 \\ 0 &\leq z \leq 1 \end{aligned}$$



Sol

$$\oint_S \vec{F} \cdot \hat{n} \, ds = \iiint_V (\text{div } \vec{F}) \, dv \quad \text{--- (1)}$$

$$\text{div } \vec{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} \quad \text{--- (2)}$$

$$= \frac{\partial}{\partial x} (2xy) + \frac{\partial}{\partial y} (2ye^z) + \frac{\partial}{\partial z} (x \sin z) \quad \text{--- (3)}$$

$$= 2y + 2e^z + x \cos z \quad \text{--- (4)}$$

03

put in (1)

03

(1) $\Rightarrow$

$$\iiint_S \vec{F} \cdot \hat{n} \, ds = \iiint_T (2y + 2e^z + x \cos z) \, dV$$

(5)

$$\begin{aligned} 0 &\leq x \leq 1 \\ 0 &\leq y \leq 1 \\ 0 &\leq z \leq 1 \end{aligned} \quad \left(\begin{array}{c} \text{from figure} \end{array} \right)$$

$$dV = dx \, dy \, dz \quad (6)$$

(5)

 $\Rightarrow$

$$\iiint_S \vec{F} \cdot \hat{n} \, ds = \int_0^1 \int_0^1 \int_0^1 (2y + 2e^z + x \cos z) \, dx \, dy \, dz$$

(7)

$$= \int_0^1 \int_0^1 \left(2yx + 2e^z x + \frac{x^2}{2} \cos z \right) \Big|_{x=0}^{x=1} dy \, dz$$

(8)

$$= \int_0^1 \int_0^1 \left[2y(1-0) + 2e^z(1-0) + \left(\frac{1^2}{2} - \frac{0^2}{2} \right) \cos z \right] dy \, dz$$

(9)

$$= \int_0^1 \int_0^1 \left(2y + 3e^z + \frac{1}{2} \cos z \right) dy dz \quad (10)$$

$$= \int_0^1 \left(\cancel{2} \frac{y^2}{\cancel{2}} + 3e^z y + \frac{1}{2} \cos z y \right) \bigg|_{y=0}^{y=1} dz \quad (11)$$

$$= \int_0^1 \left(1 + 3e^z + \frac{1}{2} \cos z \right) dz \quad (12)$$

$$= \left(z + 3 \frac{e^z}{1} + \left(\frac{+1}{2} \sin z \right) \right) \bigg|_{z=0}^{z=1} \quad (13)$$

$$\boxed{\int_S \vec{F} \cdot \hat{n} \, dS = 1 + 3(e-1) + \frac{1}{2} (\sin 1)} \quad (14)$$

Ex

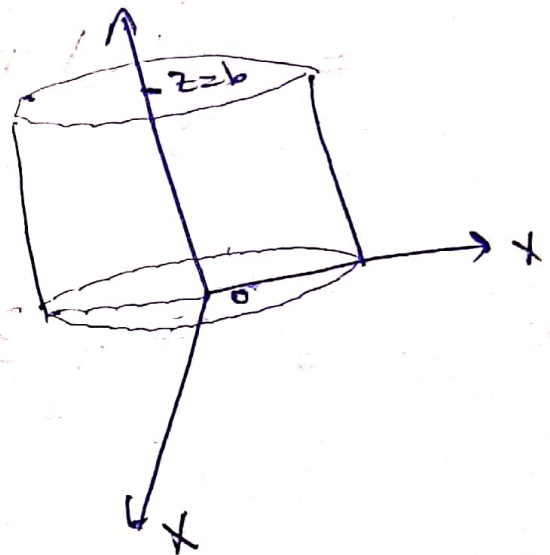
(05)

Evaluate

$$\iint_S (x^3 dy dz + x^2 y dz dx + x^2 z dx dy)$$

where S is closed surface consisting of cylinder $x^2 + y^2 = a^2$ ($a \leq z \leq b$) and circular disk $z=0$ and $z=b$.

Sol



or

$$\iiint_S P dy dz + Q dz dx + R dx dy$$

$$= \iiint_T \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} \right) dx dy dz$$

(B)

02

From (1) & (3)

(06)

$$\Rightarrow P = x^3, \quad Q = x^2y, \quad R = x^2z$$

$$\frac{\partial P}{\partial x} = 3x^2, \quad \frac{\partial Q}{\partial y} = x^2, \quad \frac{\partial R}{\partial z} = x^2$$

$$\Rightarrow \operatorname{div} \vec{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} = 3x^2 + x^2 + x^2 = 5x^2 \quad (10)$$

03 Triple integral in cylindrical coordinates
 $(x, y, z) \rightarrow (r, \theta, z)$

$$x = r \cos \theta \quad (11)$$

$$y = r \sin \theta \quad (12)$$

$$z = z \quad (13)$$

$$dv = dx dy dz = r dr d\theta dz \quad (14)$$

04

$$\int \int_S x^3 dy dz + x^2 dx dz + x^2 z dx dy$$

(3)

$$= \int \int \int_T (5x^2) dv \quad (15)$$

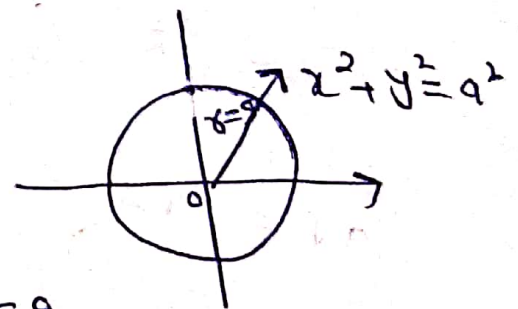
(67)

$$= \int \int \int_{z=0}^{z=b} 5(r \cos \theta)^2 r \, dz \, d\theta \, dr \quad (16)$$

$$= \int \int \int_{z=0}^{z=b} (5 r^3 \cos^2 \theta) \, dz \, d\theta \, dr \quad (17)$$

$$= \int \int \left(5 r^3 \cos^2 \theta \, z \bigg|_{z=0}^{z=b} \right) d\theta \, dr \quad (18)$$

$$= 5b \int_{\theta=0}^{2\pi} \int_{r=0}^{r=a} r^3 \cos^2 \theta \, dr \, d\theta \quad (19)$$



$$= 5b \int_{\theta=0}^{\theta=2\pi} \left[\frac{r^4}{4} \cos^2 \theta \right]_{r=0}^{r=a} d\theta$$

(20)

(08)

$$= \frac{5ba^4}{4} \int_0^{2\pi} (\cos^2 \theta) d\theta$$

$$= \frac{5ba^4}{4} 2\pi$$

Exercise

Q1 Find Flux of

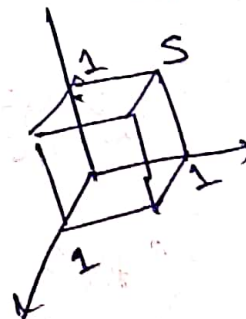
$$\vec{F} = xy\hat{i} + yz\hat{j} + xz\hat{k}$$

outward through the surface of cube

$$0 \leq x \leq 1$$

$$0 \leq y \leq 1$$

$$0 \leq z \leq 1$$



Q2 Evaluate surface integral $\iint_S \vec{F} \cdot \hat{n} \, ds$
by divergence theorem.

(i) $\vec{F} = [x^2, 0, z^2]$, S is the box

$$\begin{aligned} -1 &\leq x \leq 1 & (x \in [-1, 1]) \\ -1 &\leq y \leq 1 & (y \in [-1, 1]) \\ 0 &\leq z \leq 2 \end{aligned}$$

(ii) $\vec{F} = [e^x, e^y, z]$, S is the box

$$\begin{aligned} -1 &\leq x \leq 1 \\ -1 &\leq y \leq 1 \\ -1 &\leq z \leq 1 \end{aligned}$$

(iii) $\vec{F} = [\sin y, \cos x, \cos z]$, S is
the surface of

$$x^2 + y^2 \leq 4, \quad 0 \leq z \leq 2.$$

(As Example #02).